

Boletín de Docker Compose (soluciones detalladas)

1. Servidor web estático con Nginx

Estrutura de directorios

```
web-compose/  
├─ docker-compose.yml  
└─ index.html
```

index.html

```
<!DOCTYPE html>  
<html lang="gl">  
<head>  
  <meta charset="UTF-8" />  
  <title>Docker Compose</title>  
</head>  
<body>  
  <h1>Benvido a Docker Compose!</h1>  
  <p>Esta é unha páxina estática servida por Nginx dentro dun contedor.</p>  
</body>  
</html>
```

docker-compose.yml

```
version: "3.9"  
services:  
  web:  
    image: nginx:alpine  
    container_name: web_nginx  
    ports:  
      - "8080:80"  
    volumes:  
      - ./index.html:/usr/share/nginx/html/index.html:ro
```

Comandos

```
# Entrar no directorio  
cd web-compose  
  
# Arrincar en segundo plano  
docker compose up -d
```

```
# Proba en navegador: http://localhost:8080

# Ver logs (opcional)
docker compose logs -f

# Deter e limpar
docker compose down
```

2. Aplicación Flask con base de datos MySQL

Estrutura de directorios

```
flask-mysql/
├── docker-compose.yml
└── app/
    ├── app.py
    ├── requirements.txt
    └── Dockerfile
```

app/app.py

```
from flask import Flask
import os, mysql.connector

app = Flask(__name__)

@app.route('/')
def index():
    db = mysql.connector.connect(
        host="db",
        user="root",
        password=os.environ.get("MYSQL_ROOT_PASSWORD"),
        database="demo"
    )
    cursor = db.cursor()
    cursor.execute("SELECT 'Docker Compose con Flask e MySQL funcionando!'")
    return cursor.fetchone()[0]

if __name__ == "__main__":
    app.run(host="0.0.0.0", port=5000)
```

app/requirements.txt

```
Flask==3.0.0
mysql-connector-python==9.1.0
```

app/Dockerfile

```
FROM python:3.12-slim
WORKDIR /app
COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt
COPY . .
CMD ["python", "app.py"]
```

docker-compose.yml

```
version: "3.9"
services:
  web:
    build: ./app
    container_name: flask_app
    ports:
      - "5000:5000"
    environment:
      - MYSQL_ROOT_PASSWORD=1234
    depends_on:
      - db

  db:
    image: mysql:8.0
    container_name: mysql_db
    environment:
      MYSQL_ROOT_PASSWORD: 1234
      MYSQL_DATABASE: demo
    volumes:
      - db_data:/var/lib/mysql
    restart: always

volumes:
  db_data:
```

Comandos

```
# Ir ao directorio do exercicio
cd flask-mysql

# Arrincar en segundo plano
docker compose up -d
```

```
# Proba no navegador
# http://localhost:5000

# Ver logs (opcional)
docker compose logs -f web
docker compose logs -f db

# Deter todo e LIMPAR volumes
docker compose down -v
```